

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{array} \right] \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 4 \times \text{Row 1}}]{\substack{\text{Row 3} = \text{Row 3} \\ + 3 \times \text{Row 1}}} \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & 4b_1 + b_2 \\ 0 & 7 & 5 & 3b_1 + b_3 \end{array} \right] \xrightarrow[\substack{\times 1/2}]{\text{Row 2} = \text{Row 2}} \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & 2b_1 + b_2/2 \\ 0 & 7 & 5 & 3b_1 + b_3 \end{array} \right] \\
 & \xrightarrow[\substack{- \text{Row 2}}]{\text{Row 3} = \text{Row 3}} \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & 2b_1 + b_2/2 \\ 0 & 0 & 0 & b_1 + b_3 - b_2/2 \end{array} \right]
 \end{aligned}$$

Verification of whether the final matrix is in echelon form

- Let $b_1 + b_3 - b_2/2 = 0$. Then the third row is zero row. The first two rows are non-zero. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.
Let $b_1 + b_3 - b_2/2 \neq 0$. Then, all the 3 rows in the matrix are non-zero. $\therefore$ The condition that all the non-zero rows must be above the zero rows is vacuously True.
- Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 7 in Column 2
Possible Leading Entry of Row 3 = $b_1 + b_3 - b_2/2$ of Column 4
 $\therefore$ The condition that for each ^{row the} leading entry is in a column to the right of the columns of the leading entries of all rows above it is True.
- Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that for each pivot column, all the elements below the pivot position must be all zero is True.
 $\therefore$ The matrix $\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & 2b_1 + b_2/2 \\ 0 & 0 & 0 & b_1 + b_3 - b_2/2 \end{array} \right]$ is in echelon form $\square$